

Ismael Sallami Moreno

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Expected graduation: Jun. 2027 (final year) | **Available:** 13–17 week full-time internship from Jul. 2027 | EU citizen

EDUCATION

University of Granada

Granada, Spain

Double Degree in Computer Science and Business Administration

Sep. 2022 – Jun. 2027 (expected)

- **Algorithms & Data Structures:** public catalogue of over 180 C++ implementations organized by pattern, from graphs and dynamic programming to concurrency primitives; CI compiles and runs every reference implementation on every push.

EXPERIENCE

The Mifos Initiative

Remote

Google Summer of Code 2026 Contributor

May 2026 – Aug. 2026

- Integrated OpenSPP as a fourth Digital Public Good into Mifos Gazelle's open-source Kubernetes deployment tool and wrote the Helm chart from scratch, since none existed upstream; setup dropped from a manual docker-compose to one command. 20 merged pull requests, over 5,000 lines across 95 file changes, in 12 weeks.
- Ran the full stack of roughly 60 pods and 4 Digital Public Goods on one 16 GB node and found that CPU requests exhaust before memory, with payment pods stuck Pending at 98% of the node's requests; scaling idle web UIs to zero settled at 86%.
- Rewrote a health check that passed on an empty deploy, since a bare Odoo also answers 200, so it now asserts the module is genuinely installed. Wrote 29 unit tests for the payment connector and an 86-check Bash harness over the chart render and config consistency.
- Built the payment path from OpenSPP through Payment Hub EE and Mojaloop to Apache Fineract, verified against the Fineract audit trail after the orchestrator returned COMPLETED for transfers that moved no money: 5 of 5 subsidies credited, idempotent on re-run. A second act scored 5 farm households on a 5-indicator OpenSPP scorecard into 3 risk bands and 5 loan amounts.

Cátedra GPU Solutions, University of Granada

Granada, Spain

Undergraduate Research Scholar

May 2026, resuming Oct. 2026

- Selected for a funded six-month placement on energy efficiency in AI infrastructure, on the GPU-Lab-UGR cluster. Paused for Google Summer of Code; it becomes the basis of the undergraduate thesis.

Golden Moments Worldwide Ltd

Cork, Ireland

Digital Marketing & E-Commerce Intern (Erasmus+)

Jul. 2026 – Sep. 2026

- Shipped front-end changes in HTML, CSS and JavaScript, analyzed traffic with Ahrefs, defined the SEO strategy and handled partner negotiation and customer support, 35 hours a week entirely in English.

PROJECTS

Concurrency & Distributed Systems | C++11, MPI, OpenMPI, Real-Time Schedulers

2025

- Wrote over 8,000 lines across more than 30 programs solving producer-consumer, readers-writers, cigarette smokers and dining philosophers three times over: first with semaphores, then Hoare monitors, then MPI message passing.
- Reproduced deadlock on purpose in one version of dining philosophers, fixed it with an arbitrator process in another, and timed cyclic real-time executives against period and worst-case execution time.

Ansible Infrastructure Lab | Ansible, Prometheus, Grafana, JMeter, Rocky Linux

2025

- Automated provisioning of a 3-node Rocky Linux cluster with idempotent playbooks that pick Apache or Nginx from the inventory group, so adding a web server is one group and one vars file rather than a new conditional.
- Hardened SSH in three steps so each node ends more locked down than it started while a second run still changes nothing. Prometheus scrapes 3 targets every 5 seconds into Grafana, with a JMeter plan exercising Basic then JWT auth.

CEC2017 Metaheuristic Benchmark | C++17, Python, CMake

2026

- Benchmarked a bio-inspired optimizer across nearly 17,000 experiments (6 algorithms, 30 functions, 3 dimensions, 31 runs); a memetic Solis-Wets hybrid improved convergence 7–8 orders of magnitude (Wilcoxon $p < 0.001$) and ranked 4th of 8 against published CEC2017 baselines.
- Found a budget-truncation bug that silently dropped the final optimization milestone in 50-D runs and restored correctness across the whole 6-algorithm suite.

TECHNICAL SKILLS

Languages: C++, Python, C, Java, JavaScript, Bash, SQL

Infrastructure: Kubernetes, Helm, Docker, Ansible, Prometheus, Grafana, CircleCI, Linux (Arch, Rocky)

Fundamentals: Data Structures & Algorithms, Distributed & Parallel Systems, Site Reliability Engineering, Networking, Machine Learning

Spoken: English (professional working proficiency), Spanish (native)